Weekly Literature Review

Period: Nov 04 – Nov 11, 2026

This Week in Brief

The week's biggest news is that two leading beyond-LDL targets failed in outcome trials: pelacarsen markedly lowered Lp(a) without reducing events, and ziltivekimab suppressed IL-6 and CRP with neutral MACE, clouding the HERMES program in HFpEF. Physiology-preserving and target-driven strategies fared better: left bundle-branch pacing more than halved the composite of death, HF hospitalization, or pacing-induced cardiomyopathy versus RV pacing in AV block, and a systolic target below 120 reduced events after ischemic stroke. In transplant recipients at least a year out, dapagliflozin improved eGFR slope, though only on a short-term surrogate endpoint.

Cardiology

A heavy trial week. Two leading beyond-LDL hypotheses took major hits: pelacarsen lowered Lp(a) without reducing events, and ziltivekimab suppressed IL-6 and CRP with neutral outcomes. The positive results came from physiology-preserving and target-driven strategies: left bundle-branch pacing over RV pacing, a systolic target below 120 after stroke, and SGLT2 inhibition in heart transplant recipients. Incretin trials pushed further into HFpEF hemodynamics and obesity CV safety.

Pelacarsen and Cardiovascular Outcomes in Patients with High Lipoprotein(a)

N Engl J Med · Cho L, Nicholls SJ, Frndak S, et al. · Placebo-controlled RCT, 8,350 patients · DOI

Area: Lipoprotein(a) lowering, secondary prevention

Design: Patients with established cardiovascular disease and Lp(a) of at least 70 mg/dL were randomized to monthly subcutaneous pelacarsen 80 mg or placebo. Primary endpoint was CV death, nonfatal MI, nonfatal stroke, or urgent coronary revascularization; median follow-up 4.7 years.

Results: Median baseline Lp(a) was 236 nmol/L; at 12 months it was 64.4 nmol/L with pelacarsen vs 233.4 nmol/L with placebo. The primary endpoint occurred in 12.3% vs 12.1% (HR 1.02, 95% CI 0.90 to 1.15, P = 0.76). In patients with Lp(a) of at least 90 mg/dL, rates were 12.3% vs 12.3% (HR 1.00; 95% CI 0.87 to 1.15). Serious adverse events: 37.3% vs 41.1%.

Context: The first Lp(a) outcomes trial is flatly neutral despite driving Lp(a) from 233.4 to 64.4 nmol/L over nearly 5 years. The null result in the higher-Lp(a) subgroup weakens the argument that the reduction was too small. Mendelian randomization still supports causality, so the siRNA outcome trials with deeper suppression, olpasiran and lepodisiran, now carry the hypothesis. For now, Lp(a) remains a risk-enhancer that should prompt intensification of proven therapies, not a validated treatment target.

Interleukin-6 Inhibition with Ziltivekimab for Atherosclerotic Disease

N Engl J Med · Ridker PM, Lincoff AM, Marx N, et al. · Double-blind RCT, 6,376 patients · DOI

Area: Anti-inflammatory therapy, ASCVD with CKD

Design: Adults with atherosclerotic disease, chronic kidney disease and hsCRP of at least 2 mg/L were randomized to ziltivekimab 15 mg subcutaneously monthly or placebo. Primary endpoint was 3-point MACE; median follow-up 33 months.

Results: At 24 months, the placebo-adjusted median change was -76 percentage points for free IL-6 and -64 for hsCRP. The primary endpoint occurred in 17.6% vs 17.7% (HR 0.99; 95% CI 0.88 to 1.11; P = 0.84). Serious adverse events were 54.6% vs 54.8%, but serious infections were 20.4% vs 17.9%.

Context: Deep IL-6 and CRP suppression did nothing for MACE (HR 0.99) in the population thought most likely to benefit, along with a small excess of serious infections. This undercuts the genetic IL-6 receptor data and the expectation of a CANTOS-like benefit from moving downstream of IL-1β. Together with the null CLEAR SYNERGY colchicine result, it suggests elevated CRP may be more marker than modifiable mediator, at least in advanced CKD with high competing risk. The HFpEF program with ziltivekimab, HERMES, now looks less certain.

Left Bundle-Branch Pacing versus Right Ventricular Pacing for Atrioventricular Block

N Engl J Med · Qian Z, Jiang L, Hou X, et al. · Open-label RCT, 702 patients · DOI

Area: Conduction system pacing, AV block

Design: Patients with atrioventricular block were randomized to left bundle-branch pacing or right ventricular pacing. Primary composite was all-cause death, heart failure hospitalization, or pacing-induced cardiomyopathy. The trial was stopped early at a prespecified interim analysis; median follow-up 35.1 months.

Results: In the modified ITT population of 685, primary events occurred in 16 vs 35 patients, with 4-year cumulative incidence of 8.5% vs 19.2% (HR 0.41; 95% CI 0.23 to 0.75; P = 0.003). Serious adverse events unrelated to implantation were 11.8% vs 9.3% (P = 0.32).

Context: This is the first outcomes trial showing left bundle-branch pacing beats RV pacing in AV block, cutting the composite by more than half. Prior support was observational or physiologic. Caveats temper the effect size: the trial was open-label, stopped early on only 51 events, run at expert centers, and pacing-induced cardiomyopathy is a soft component likely driving much of the difference. Still, it will push conduction system pacing toward default for patients expected to pace heavily. Long-term lead performance and extractability remain the open questions.

Intensive Blood-Pressure Control in Patients with Previous Ischemic Stroke

N Engl J Med · Sampaio Silva G, Machline-Carrion MJ, Santo K, et al. · Open-label RCT, 4,368 patients · DOI

Area: Blood pressure targets after stroke

Design: Brazilian adults with elevated BP and prior ischemic stroke or TIA were randomized to a systolic target below 120 vs below 140 mm Hg. Primary composite was CV death, MI, stroke, or hospitalization for unstable angina or heart failure; median follow-up 3.9 years.

Results: Achieved mean SBP was 122.0 vs 135.2 mm Hg. The primary outcome occurred in 10.9% vs 13.5% (3.25 vs 4.11 events per 100 person-years; HR 0.79; 95% CI 0.67 to 0.94; P = 0.007). The composite renal outcome was more common with intensive treatment; overall serious adverse events were similar.

Context: Targeting SBP below 120 after ischemic stroke or TIA reduced major CV events by about a fifth, filling the gap left by SPRINT, which excluded stroke. The companion OPTIMAL-DIABETES trial from the same investigators was null. That echoes ACCORD-BP but conflicts with the positive BPROAD trial in diabetes, so the diabetes question remains unsettled. Open-label design and the renal signal warrant monitoring, but this supports intensive targets in post-stroke patients who tolerate them.

Dapagliflozin for Renal Protection in Heart Transplant Recipients: A Randomized Clinical Trial

Circulation · Double-blind RCT, 430 patients · Heart transplantation, nephroprotection

In recipients at least one year post-transplant, dapagliflozin 10 mg vs placebo improved the 12-month chronic eGFR slope by 1.7 mL/min/1.73 m² (95% CI 0.1 to 3.3; p = 0.03), without excess infections. The endpoint is a surrogate with short follow-up and no hard renal or graft outcomes.

Also notable

- Discontinuing Statins in Older Adults without Cardiovascular Disease (*N Engl J Med*). STREAM: in 1,880 adults aged 70+ on primary-prevention statins, discontinuation met noninferiority, yet primary events were 12.7% versus 9.2% with continuation.
- Long-Term Survival After Initial Invasive or Conservative Management of Stable Coronary Disease in ISCHEMIA-EXTEND (*JAMA*). ISCHEMIA-EXTEND: 11-year all-cause mortality was 25.8% invasive versus 26.9% conservative (adjusted HR 0.97), confirming no long-term survival benefit of routine revascularization.
- Cangrelor versus crushed ticagrelor in acute myocardial infarction complicated by cardiogenic shock (DAPT-SHOCK-AMI): a randomised, double-blind trial (*Lancet*). DAPT-SHOCK-AMI: in AMI cardiogenic shock, cangrelor inhibited platelets better than crushed ticagrelor but missed 30-day noninferiority (38% vs 41%).

AI in Medicine

This week's AI work is moving from model accuracy to prospective and randomized testing, with sobering results. An AI-ECG–guided AF screening trial produced a statistically fragile gain, and a year-long generative AI chatbot trial showed no mental health benefit. Deployment studies show tools can find more disease, from smartwatch hypertension to inpatient delirium, but they have not yet shown that finding it changes outcomes.

Leveraging AI-ECG Technology for Early Notification and Tracking of AF Development (LATENT): A Randomized Controlled Trial

Circulation · Lee CC, Lin WY, Liu WT, et al. · Single-center open-label RCT, 14,665 adults · DOI

Area: AI-ECG for atrial fibrillation detection

Design: Adults aged 40–85 without prior AF/AFl were randomized to an AI-ECG–guided strategy or usual care. AI-flagged high-risk intervention patients were offered Holter monitoring for up to 7 days, and matched controls received identical monitoring. Primary endpoint: new AF, AFl, or pre-AF atrial findings within 90 days.

Results: AI flagged 84 (1.1%) intervention and 78 (1.1%) control participants as high risk; 48 (57.1%) of flagged intervention patients were monitored. The primary outcome occurred in 54 (0.74%) vs 35 (0.48%) participants (OR 1.54, 95% CI 1.01-2.37; P=0.046), and reclassifying a single event moved P to 0.057. Among high-risk patients, yield was 32.1% vs 2.6%. AF/AFl detection (0.3% vs 0.2%) and antiarrhythmic initiation at 180 days (2.3% vs 2.4%) did not differ.

Context: AI-ECG picks out a small sinus-rhythm group with very high arrhythmia yield on Holter, but at the population level the benefit was fragile. It was driven mainly by pre-AF findings such as frequent PACs rather than AF itself. Prior AF screening trials have shown that more detection does not reliably translate into fewer strokes, and this trial measured neither anticoagulation nor outcomes. Only about 1% were flagged, and just over half of those were monitored. This is a proof of concept for targeted surveillance, not a reason to adopt it.

Generative Artificial Intelligence and Maternal Health Outcomes in Pregnant and Postpartum Individuals: A Randomized Clinical Trial

JAMA Netw Open · Norell EM, Yue X, Mane H, et al. · RCT with active control, 400 participants · DOI

Area: Patient-facing generative AI chatbot, maternal health

Design: Pregnant or postpartum (infant younger than 6 months) individuals from racial and ethnic minoritized groups were recruited online across the US. They were randomized 1:1 to access to a generative AI maternal health chatbot or an active control (monthly children's books). Primary endpoints were GAD-7 and PHQ-9 at 12 months.

Results: The chatbot did not change anxiety (adjusted mean difference 0.10; 95% CI, -0.77 to 0.96) or depression (-0.23; 95% CI, -1.05 to 0.59). Emergency department visits, postpartum care, contraception use and sleep duration also did not differ. Among those pregnant at enrollment, more self-reported pregnancy complications appeared at 12 months but not in longitudinal analyses. Higher engagement correlated with longer sleep but also higher anxiety in exploratory analyses.

Context: In a rare year-long randomized test of a patient-facing generative AI tool, simply giving patients chatbot access did nothing for maternal anxiety or depression. That is a useful counterweight to engagement-driven enthusiasm. The exploratory engagement associations cut in both directions, a reminder that usage-outcome correlations are confounded. Limitations include an access-based intervention, self-reported outcomes and a model version that will age quickly. Still, the lesson holds: patient-facing LLM tools need outcome trials before anyone assumes benefit.

Opportunistically Detecting Signs of Hypertension on a Consumer Smartwatch

Nature · Prospective multicenter validation, 196 participants · Smartwatch hypertension screening

An AI model using smartwatch PPG and accelerometry, without cuff calibration, was compared against 24-

hour ambulatory BP over 7 days. It detected hypertension with sensitivity 65.8%, specificity 90.0% and PPV 80.6%, versus office BP sensitivity 55.3%. The cohort was small, untreated and free of CVD, and the model missed a third of cases.

AI-Identified Reasons for Nonescalation of Lipid-Lowering Therapy for Secondary Prevention

JAMA Cardiol · Retrospective cohort, 28,113 ASCVD patients · LLM extraction of lipid therapy inertia

An LLM read clinic notes from ASCVD patients with LDL-C at or above 70 mg/dL, with F1 scores of 0.80 to 0.98 against annotators. Therapy was not escalated in 88.7% of notes, most often because of patient-reported adverse effects (12.6%). This is a single health system, and the method only captures reasons clinicians documented.

External Validation of a Machine Learning Model for Delirium Risk

JAMA Netw Open · Quality improvement external validation, 33,095 admissions · ML delirium risk model, live deployment

A multimodal NLP delirium model deployed at a second hospital without retraining achieved an AUROC of 0.91. Monthly delirium detection rose from 9.23% to 19.59% after deployment. However, PPV was only 25.9%, and the uncontrolled pre-post design cannot show patient benefit.

AI in Medical Education

This week's AI-in-education work was thin. The most rigorous study, a multicenter randomized trial of AI feedback on Doppler interpretation, found that AI helps trainees and experienced clinicians about equally, but trainees were more likely to follow wrong advice. A national survey shows that medical school policy has not caught up with student use. Controlled evidence that AI tools actually improve learning remains scarce.

AI feedback for clinician assessment of umbilical and middle cerebral artery Doppler ultrasound: a multicenter randomized trial

Am J Obstet Gynecol MFM · Bashir Z, Wong CK, Lin M, et al. · Multicenter RCT, 96 clinicians · DOI

Area: AI decision support and automation bias in obstetric ultrasound

Design: Obstetrics residents and specialists at five Danish hospitals were randomized 1:1 to AI visual and text feedback or none. Each interpreted 40 static umbilical and middle cerebral artery Doppler vignettes (gate placement, waveform adequacy). The primary outcome was accuracy against expert consensus. There was no real-time scanning and no unassisted retest.

Results: Accuracy was 75.0% with AI feedback vs 62.3% without (mean difference 12.7%; 95% CI, 8.2-17.1; $p < 0.001$). The gain was similar in novices (70.0% vs 57.9%; 95% CI 2.4-21.7) and experienced clinicians (78.0% vs 65.0%; 95% CI 5.5-20.5). Novices followed incorrect AI feedback more often than experienced clinicians (22.3% vs 15.8%; P = .01). AI alone beat unaided novices, while experienced clinicians with AI beat AI alone.

Context: AI feedback raised Doppler interpretation accuracy by about 13 points regardless of experience. Experience still mattered when the AI was wrong: novices were likelier to accept incorrect output. This fits the pattern from mammography and colonoscopy, where AI helps in bounded perceptual tasks, and contrasts with the GPT-4 vignette trial, where model access did not improve physicians' open-ended diagnostic reasoning. The key gap for education is that this measured assisted performance, not learning. Whether trainees who practice with such feedback get better or worse unaided remains untested.

Policy, ethical, and practical considerations of artificial intelligence use in the internal medicine clerkship: results of a survey of clerkship directors

Acad Med · National survey, 114 clerkship directors · AI policy in internal medicine clerkships

Of responding US internal medicine clerkship directors (80.9% response), 39.5% reported no formal AI policy and 34.2% did not know whether one existed. Although 73.7% would encourage AI use for at least one clinical skill, only 6.1% had institutional firewall-protected tools. The data are self-reported and from 2024.

Evaluating an AI-Powered Patient Simulator for Communication Skills Training: One-Year Randomized Controlled Trial

J Med Internet Res · Nonblinded RCT, 50 medical students · Generative AI patient simulator for communication skills

Over one academic year at a UK school, students practicing with a chatbot patient simulator showed descriptively larger OSCE communication gains than conventional training, but the between-group difference was not statistically significant. Motivation and behavioral engagement predicted improvement. The trial was small, single-site and underpowered.

Longevity Medicine

No human geroprotector trials this week, and the human data pushed back on popular longevity claims. A large transplant registry found no survival benefit from chronic rapamycin. Whole-blood NAD did not decline with healthy aging. The most novel mechanistic work was a mouse senolytic, vanadyl sulfate, which exploits a redox weakness specific to senescent cells. The rest was mostly preclinical biology, epigenetic-clock variations and Lancet Healthy Longevity reviews on exercise.

Rapamycin and human longevity among kidney transplant recipients surviving the first year

Geroscience · Registry cohort, 11,474 kidney transplant recipients · mTOR inhibition and human longevity

Among 1-year kidney transplant survivors (4885 on rapamycin, 6589 not; median follow-up 11 years), propensity-weighted median survival age was 64.6 vs 64.3 years (95% CI for difference -1.6, 2.0 years). Caveats: observational, immunosuppressive dosing, and a multimorbid population far from geroprotection use.

Whole-blood NAD levels track cardiovascular disease but not chronological age

Nat Metab · Cross-sectional cohort, 1,068 adults · NAD metabolism and aging biomarkers

Across 1,340 measurements, whole-blood NAD did not decline with age in healthy adults; it was stable in males and rose in females. It was lower in cardiovascular disease and hypertension. This undercuts the

age-related NAD decline rationale for supplements, but the design is cross-sectional and whole blood may not reflect tissue NAD.

Altered protein cysteine reactivity in senescent cells is a vulnerability for senolysis to ameliorate aging

Nat Aging · Preclinical mechanistic and mouse lifespan study · Senolytics

Oral vanadyl sulfate, a mineral supplement, selectively killed senescent cells and extended lifespan in male mice. Senescent cells accumulate glutathione disulfide because glutathione reductase is downregulated, and vanadyl sulfate exploits this to drive protein misfolding. Mouse-only, male-only, with no human data.

General Medicine

Three themes this week. First, incretin indications keep widening beyond weight. Retatrutide improved knee osteoarthritis pain alongside about 20% weight loss, and oral orforglipron reduced sleep apnea severity whether or not patients used PAP. A Nature Medicine trial found resistance training preserved lean mass and strength on tirzepatide. Second, more intensive treatment is not always better. In older, frail adults with kidney failure, conservative management did not cost quality-adjusted life compared with preparing for dialysis. In type 2 diabetes, OPTIMAL-DIABETES found that a systolic target below 120 did not significantly reduce cardiovascular events, unlike SPRINT in non-diabetic patients. A pharmacist-led deprescribing bundle produced only modest gains that faded once the stewardship stopped. Third, preventive-care erosion has measurable harm: refusal of newborn vitamin K has doubled in the US and is tied to more intracranial bleeding. Separately, first-trimester doxycycline was associated with more pregnancy loss than amoxicillin or macrolides.

Preparing for responsive management versus renal dialysis in older people living with kidney failure and frailty or multimorbidity in the UK (Prepare for Kidney Care): a two-arm, superiority, parallel-group, open-label, pragmatic, phase 3, individually randomised controlled trial

Lancet · Pragmatic open-label RCT, 445 patients · Kidney failure in older adults

In older adults with kidney failure and frailty or multimorbidity, conservative management versus dialysis preparation gave no significant QALY difference (1·354 vs 1·472; adjusted difference -0·147, 95% CI -0·350 to 0·056).

Retatrutide Once Weekly in Adults with Knee Osteoarthritis Pain and Obesity

N Engl J Med · Two phase 3 RCTs, 1,019 patients · Knee osteoarthritis and obesity

Retatrutide 9 mg improved WOMAC pain by -3.96 versus -2.12 with placebo at 68 weeks in TRIUMPH-4, with weight loss of -20.0% to -23.7%. Serious adverse events were more common.

Orforglipron for Obstructive Sleep Apnea: Results From 2 Randomized Clinical Trials (ATTAIN-OSA)

JAMA · Two phase 3 RCTs, 712 patients · Obstructive sleep apnea

Oral orforglipron reduced the apnea-hypopnea index versus placebo at 52 weeks (differences -13.2 and -12.0 events/h, without and with PAP). GI events occurred in 68.0% vs 37.4% in trial 1.

Intensive Blood-Pressure Lowering in Patients with Type 2 Diabetes

N Engl J Med · Open-label RCT, 9476 patients · Blood pressure targets in type 2 diabetes

In OPTIMAL-DIABETES, a systolic target below 120 versus below 140 mm Hg did not significantly reduce major cardiovascular events over 4.0 years (6.0% vs 6.7%; HR 0.90, 95% CI 0.77 to 1.06).

Nonreceipt of Newborn Vitamin K and Risk of Bleeding

JAMA · Retrospective cohort, 5 240 780 infants · Newborn vitamin K prophylaxis

US newborns not receiving vitamin K rose from 3.1% to 6.8% (2017 to 2025). Nonreceipt carried higher intracranial bleeding risk (ARR 2.43, 95% CI 2.26-2.62).

Verification Notes

- 497 candidate papers screened from PubMed; 21 selected (7 with a full write-up).
- Every selected PMID, title, author list and journal is taken directly from PubMed. Design and Results lines are drawn from the abstract and their numbers checked against it automatically. Context lines reflect the model's knowledge of the prior literature and are not machine-verified.
- AI in Medicine, PMID 42988369: the write-up cites 7, not found verbatim in the PubMed abstract. Check against the paper.

Selected and written by claude-opus-5-5 from PubMed records added Nov 04 – Nov 11, 2026.